

Course Code: CSA1510

(18)

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1. unified Big data Architecture for a multinational Retail organization.

A Multinational retailer collects customer data from websites, mobile apps, pos systems, stores, loyalty programs, and social media. Since the data is stored separately, it is difficult to understand customer behavior and provide personalized services.

1. PROBLEMS

- * Data is stored in different system and formats.
- * Online and offline customer data is not properly connected.
- * Large amounts of data are generated continuously
- * Traditional analytics are not real-time.
- * Customers receive generic recommendations.

2. PROPOSED ARCHITECTURE

Data Sources → Kafka → Data Lake → Spark
→ Customer 360 → ML Analytics → Applications

Data Sources: website, mobile app, pos, loyalty cards, social media, and stores.

Kafka: collects customer activities in real time.

Data Lake: Stores large volumes of customer and transaction data.

Spark: cleans, processes, and analyzes the data.

Customer 360: creates one complete customer profile using purchase history, browsing behaviour, store visits, and preferences.

ML Analytics: Generates personalized product recommendations and offers.

3. Real-time Analytics

for example, when a customer searches for a laptop, kafka captures the activity, spark processes it, and the recommendation system immediately suggests laptops, accessories, or discounts.

4. Security

The system uses encryption, access control, data masking, consent management, and data governance to protect customer information.

5. Benefits

- * unified customer view
- * Real-time analytics
- * Personalized recommendations.
- * Better customer experience.
- * Increased sales and customer retention.
- * Scalable global architecture.

Conclusion:

The proposed big data architecture combines online and offline customer data into a single platform, enabling real-time analytics and personalized recommendations while improving business performance.

4. Multimedia Data Management in Big Data

Multimedia data such as videos, audio and images is large, unstructured, and generated continuously. Managing this data in a big data environment creates several challenges.

1. Challenges

Huge data size:

Videos and high resolution images require large storage capacity.

High processing requirements:

Processing multimedia files needs high CPU and memory.

Different formats:

Data may be stored as MP4, JPEG, PNG, MP3, etc.

Fast data generation:

Streaming platforms generate data continuously.

Data retrieval:

Finding & accessing specific multimedia files can be difficult.

2. Suitable Technologies

HDFS:

Stores large multimedia files across multiple machines with fault tolerance.

Cloud object Storage:

Services such as Amazon S3, Azure Blob Storage, provide Scalable Storage.

Apache Spark:

Processes large-scale multimedia metadata.

Kafka:

Handles real-time video / audio streaming data.

NoSQL databases:

MongoDB can store multimedia metadata.

CDN:

Helps deliver video and images quickly to user.

Conclusion:

A combination of distributed storage, cloud object storage, kafka, spark, NoSQL databases provides Scalable efficient management of multimedia big data.

6. Impact of Data Quality issues and data governance

Poor data quality in a large analytics platform can cause incorrect analysis, unreliable reports, and poor decisions.

1. Impact of Data Quality issues

Inconsistency: Different formats can produce incorrect results.

Duplication: Repeated records cause inaccurate calculations

Missing values: Reduce the accuracy of analytics and ML models

Poor Decisions: Incorrect data leads to wrong business strategies

Loss of trust: users may lose confidence in reports.

3. Data Governance Strategies

Data Validation: check data accuracy and format.

Data cleaning: Remove duplicates and handle missing values.

Data

Standardization: Use common data formats.

Data ownership: Assign responsibility for datasets.

Quality monitoring: continuously check data quality

Access control: Protect data from unauthorized access

Conclusion:

Good data governance improves accuracy, reliability, and trust, leading to better analytical decisions.